Computational Coupling

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Four choices decide whether a connectome beats its rewiring.

Computational coupling is the extent to which what a network computes is attributable to how it is wired. It is never measured directly — only as the gap between a network's score and the scores of surrogate networks, under some unit model, at some operating point, through some readout. This project shows that each of those four choices can flip the sign of that gap on identical data, so a coupling reported as a single signed number is under-determined by its own methods section.

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Paper: *Computational Coupling: Four Choices Decide Whether a Connectome Beats Its Rewiring* — `paper/computational_coupling_v5.pdf`

> **This version supersedes and retracts an earlier one.** Every withdrawn claim and the evidence against it is in `AUDIT.md`. See Relationship to the previous version.

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What this project asks

Connectomes contain statistically overrepresented subgraphs, and it is widely argued that these *motifs* are what make biological wiring computationally effective. The standard evidence is a comparison: score a real connectome on a task, score a rewired version of it, and credit the wiring with the difference.

That difference is the quantity we call **computational coupling**. Writing it down makes its dependencies visible:

> κ(W; N, U, ρ, R) — the standardised gap between the capacity of network `W` and the capacity of surrogates drawn from ensemble `N`, measured with unit model `U`, at operating point ρ, through readout protocol `R`.

This is the Z-score the field already reports. Naming it is useful because the name carries its four arguments. "The connectome beats its rewiring" is the claim $\kappa > 0$ — and κ is a function of four choices besides the connectome, each normally reported in a subordinate clause, if at all.

We varied all four, and each one can flip the sign of κ on identical data. A coupling reported as a single signed number is therefore under-determined by its own methods section.

The project also builds the control that the motif argument actually needs — an ensemble differing from a connectome **in its motif census and in nothing else** — because rewiring perturbs the census together with every statistic that co-varies with it, so a difference against an ordinary rewired null cannot be attributed to motifs in particular.

Choice	Setting A	Setting B
Null ensemble `N`	BCT `randmio`: weights travel with edges	Rank-reassigned: same weights, re-placed by degree ra
Unit model `U`	Graded `tanh`, never silent	LIF, silent below threshold
Operating point ρ	At criticality, $\rho = 1$	Above it, $\rho > 1$
Readout `R`	All-to-all: every neuron driven and decoded	Sensory/motor-disjoint

The fourth row is the limiting case: under an all-to-all readout the coupling is zero by construction, because no change to the wiring could have moved the number.

What is new here

1. "Degree-preserving randomisation" names two different ensembles, and they disagree (`experiments/ladder/run_unit_model_comparison.py`, `experiments/ladder/run_null_mechanism.py`). The reservoir literature reports that empirical connectomes beat degree-preserving rewirings on memory capacity — a result increasingly invoked as motivation for neuromorphic hardware. The phrase describing the control does not determine an ensemble: a weight can travel with its edge (BCT `randmio_*_connected`, what `conn2res` calls) or be re-assigned to the rewired graph afterwards. We built the second believing it was the first. Across 10 (network, unit model) cells the two give the connectome's advantage opposite **signs** in 6 and opposite **significant verdicts** in 3; against `randmio` the connectome leads in **1 of 5** networks with graded units, against rank-reassignment in **5 of 5**, median +2.13. The cause is that rank-reassignment *installs* a weight–degree association the data lacks (+0.66 median shift against +0.06), which predicts the two must agree exactly on an unweighted network — and on the one binary network in the set they agree to 0.008 capacity units. This is not a refutation of the published finding; it shows the null choice alone moves the result by more than the effect being reported.

1b. What a unit is also matters (same experiment). Graded `tanh` units are never silent; sub-threshold LIF units contribute nothing. Capacity compresses by a factor of **4.9** between the two, and **60.7%** of spiking units never fire against **0.6%** of graded ones. Our predicted mechanism — *differential* silencing — was reported as refuted against our mistaken null and is **supported** against the published one ($r = -0.58$, $p = 2.5\times10^{-5}$ over 45 cells). We report both, and the audit records why the conclusion moved.

2. The contrast also changes sign along the gain axis (`experiments/ladder/run_gain_sweep_ladder.py`). The reservoir literature reports that empirical connectomes beat degree-preserving rewirings on memory capacity, measured at criticality. We sweep the gain axis and find the contrast **changes sign** along it — in 3 of 5 networks on task accuracy, significant on both sides, swinging up to 23.4 percentage points within a network. At criticality the *unstructured* graph leads in all five. We also attempted the published memory-capacity comparison directly and did not detect the advantage, but report that as a **non-detection rather than a refutation**, because our design there resolves only an effect about half the size of the null's own capacity. A single-operating-point comparison establishes a fact about that operating point; reported without the curve it is indistinguishable from a fact about the topology.

3. A motif-preserving null ensemble (`src/nulls/swap_chain.py`). Degree-, valence- and census-matched surrogates, sampled by an annealed Metropolis chain over exact, incrementally updated motif counts. It reproduces a connectome's motif spectrum while randomising everything else, and it does *not* reproduce the held-out 4-node statistic it was not asked to preserve.

4. A structural control ladder (`src/nulls/ladder.py`). Five nested rungs, each adding exactly one structural property to the one below: density → degree sequence → valence placement → motif census → the empirical connectome. The contribution of each property is the difference between adjacent rungs, with everything else matched by construction.

5. Two controls that are usually missing (`experiments/ladder/`). A reservoir comparison is only a topology comparison if the recurrent matrix is used and if the networks are compared at the same operating point. We report the zero-recurrence control for the first, and re-run the whole ladder with every network individually recalibrated to a common firing rate for the second — matching spectral radius equalises gain, not activity level, and activity level both predicts accuracy and sets the synaptic-operation count.

6. A gap-free perturbation theory (`src/theory/transfer_perturbation.py`). The Davis–Kahan bound usually invoked for degeneracy is inapplicable to measured connectomes — their spectral gaps make it vacuous. We bound the finite-horizon input–output map instead, which needs no gap, and derive the **operational alignment** γ of a perturbation, with the falsifiable prediction that tolerance scales as $1/\gamma$. The prediction is tested against a constructed maximally-damaging perturbation, not only against random ablation.

7. An honest hardware report (`experiments/hardware/run_open_synthesis.py`). Real logic synthesis of the accompanying RTL with Yosys + ABC onto the Nangate45 open cell library, reporting only what such a flow establishes.

Relationship to the previous version of this project

This work **supersedes and retracts** the previous manuscript in this repository. The earlier version reported commercial 28 nm silicon figures that were never measured, a human microscale tier whose connectivity graph was empty, sign-attributed motif results for datasets containing no usable inhibition, p-values below the resolution of the null ensemble that produced them, and an ablation table whose own control row contradicted the conclusion drawn from it.

Every withdrawn claim and the evidence against it is documented in `AUDIT.md`. It is kept in the repository deliberately: the corrected analysis is only interpretable against a record of what it replaces. The retracted documents themselves are in `paper/superseded_v4/` and their data in `results/superseded_v4/`, moved aside and labelled, because they previously sat beside the current manuscript and its results with nothing to indicate they had been withdrawn. That includes the v4 **executive pitch dossier**, every hardware number in which is fabricated; it carries its own retraction notice because a pitch is the artefact most likely to be forwarded. There is no replacement pitch.

One finding from that audit is worth repeating here, because it is easy to reproduce by accident. In a reservoir where **every neuron is both driven and decoded**, zeroing the recurrent matrix entirely leaves benchmark accuracy unchanged — the wiring is not used. Any structural comparison run in that regime measures the input projection. The protocol used here drives a sensory subset and decodes from a disjoint motor subset, and reports the zero-recurrence control as the diagnostic that the benchmark is sensitive to topology at all.

A second finding from the audit has the same shape. Matching spectral radius across networks does **not** match the rate at which they fire, Erdős–Rényi surrogates run roughly twice as hot as the connectomes they are matched to at the same gain, and firing rate predicts accuracy. A comparison at matched gain therefore confounds topology with operating point — and, like the protocol failure, it produces entirely plausible numbers with nothing in the output to indicate a problem.

Removing that confound did not give us the result we expected. At a matched firing rate the Erdős–Rényi control still beats the connectome, by slightly *more* than at matched gain. On this benchmark connectome wiring is measurably worse than an unstructured graph of the same size and density, and every structural property that moves accuracy at all moves it downward — including the motif census itself, which costs a median 3.5 percentage points against 4.7 for the degree sequence. The motif hypothesis predicts the opposite sign. We report it that way.

Data

Network	Source	Valence available
C. elegans whole nervous system	Cook et al. 2019	yes
Drosophila central complex, antennal lobe	FlyWire v630	yes
Mouse V1 L2/3	MICrONS Pinky100	**no** — every annotated inhibitory soma is
Human, regional	HCP group consensus tractography	**no** — tractography carries no synaptic si

Task data: **Spiking Heidelberg Digits** (Cramer et al. 2022), a real spike-based benchmark, replacing the synthetic pattern tasks used previously (which saturated near 100% for every architecture and so could not discriminate between them).

Layout

```
src/motifs/fast_census.py        exact sign-attributed motif census, O(d) incremental updates
src/nulls/swap_chain.py          degree-preserving chains + the annealed motif-preserving null
src/nulls/ladder.py              the five-rung structural control ladder
src/theory/transfer_perturbation.py  gap-free bound, operational alignment, adversarial ranking
src/eval/reservoir.py            reservoir protocol (sensory/motor disjoint)
src/eval/dynamics.py             memory capacity, kernel & generalisation rank
src/eval/trained_snn.py          surrogate-gradient training of masked recurrent SNNs
src/tasks/shd.py                 Spiking Heidelberg Digits loader
experiments/                     experiment drivers (one per result)
src/analysis/                    figure, table and LaTeX-macro generation
hardware/rtl/                    Verilog sources
AUDIT.md                         record of withdrawn claims
paper/superseded_v4/             the retracted v4 manuscript, figures and pitch dossier
results/superseded_v4/           the result files those claims were drawn from
```

The listing above is the code the current paper uses. Other packages under `src/` (`architecture/`, `benchmarks/`, `learning/`, `remodeling/`, `simulation/`) and the matching directories under `experiments/` belong to v4 and are not exercised by any current result. They are kept because `AUDIT.md` cites them as the origin of specific withdrawn numbers — the energy and area constants of §A1 are literals in `hardware/sim/cen_rtl_golden_model.py`, and the synthetic benchmark of §B2 is generated by `experiments/benchmarks/run_dvs_gesture_benchmark.py` — and deleting them would make the retraction harder to check rather than easier.

Every experiment in this study

15 of 15 experiments have results on disk. Sizes are counted from each experiment's own output file when this document is built, so they cannot drift from what was actually run. The unit differs by row: a "run" is one network at one rung trained to convergence, an "evaluation" one network at one operating point probed for one metric, so the column does not sum to anything meaningful.

#	Experiment	What it asks
1	**Motif enrichment**`experiments/motifs/run_motif_significance.py`	Are the classic su
2	**Motif-preserving null ensemble**`experiments/ladder/run_null_validation.py`	Validation of the
3	**Sampler robustness**`experiments/ladder/run_null_robustness.py`	The same ensemb
4	**Protocol controls**`experiments/ladder/run_protocol_controls.py`	What does the be
5	**Structural control ladder**`experiments/ladder/run_structural_ladder.py`	L0 density to L4
6	**The ladder at matched firing rate**`experiments/ladder/run_rate_matched_ladder.py`	The same decomp
7	**Extraction check**`experiments/ladder/run_extraction_check.py`	Does taking a siz
8	**Two null ensembles, two unit models**`experiments/ladder/run_unit_model_comparison.py`	The headline exp
9	**Why the two nulls differ**`experiments/ladder/run_null_mechanism.py`	Structural descrip
10	**Spiking tuning control**`experiments/ladder/run_spiking_tuning_control.py`	Input gain x thre
11	**Gain sweep of the ladder**`experiments/ladder/run_gain_sweep_ladder.py`	The whole decom
12	**Memory-capacity sweep**`experiments/ladder/run_memory_capacity_sweep.py`	Locating the pub
13	**Intrinsic dynamics**`experiments/ladder/run_dynamics_ladder.py`	The same decomp
14	**Pruning frontier**`experiments/degeneracy/run_pruning_frontier.py`	Random, hub-tar
15	**Open-PDK synthesis**`experiments/hardware/run_open_synthesis.py`	Real logic synthes

Reproducing

```
git clone https://github.com/Ashok-clg/computational-coupling.git
cd computational-coupling
python -m venv .venv && .venv/bin/pip install -e .

# Task data (~350 MB)
mkdir -p data/raw/shd && cd data/raw/shd
curl -L -O https://zenkelab.org/datasets/shd_train.h5.gz
curl -L -O https://zenkelab.org/datasets/shd_test.h5.gz && gunzip *.gz && cd -
```

Then every experiment. This list is generated from the same source as the inventory above and the table in the manuscript, so all three agree by construction:

```
python experiments/motifs/run_motif_significance.py      # Motif enrichment
python experiments/ladder/run_null_validation.py         # Motif-preserving null ensemble
python experiments/ladder/run_null_robustness.py         # Sampler robustness
python experiments/ladder/run_protocol_controls.py       # Protocol controls
python experiments/ladder/run_structural_ladder.py       # Structural control ladder
python experiments/ladder/run_rate_matched_ladder.py     # The ladder at matched firing
↪  rate
python experiments/ladder/run_extraction_check.py        # Extraction check
python experiments/ladder/run_unit_model_comparison.py   # Two null ensembles, two unit
↪  models
python experiments/ladder/run_null_mechanism.py          # Why the two nulls differ
python experiments/ladder/run_spiking_tuning_control.py  # Spiking tuning control
python experiments/ladder/run_gain_sweep_ladder.py       # Gain sweep of the ladder
python experiments/ladder/run_memory_capacity_sweep.py   # Memory-capacity sweep
python experiments/ladder/run_dynamics_ladder.py         # Intrinsic dynamics
python experiments/degeneracy/run_pruning_frontier.py    # Pruning frontier
```

```
python experiments/hardware/run_open_synthesis.py        # Open-PDK synthesis
```

Then the statistics, figures, tables, macros and three LaTeX passes. This reports unresolved numbers, undefined references and LaTeX errors at the end, because "the build succeeded" is not evidence of anything on its own -- pdflatex exits 0 on a hard error by design:

```
./src/analysis/rebuild_paper.sh
```

`experiments/ladder/run_trained_ladder.py` is deliberately **not** in that list. It trains the recurrent weights rather than freezing them, which is the natural next experiment and is discussed as such in the paper, but it has not been run to publication standard and nothing in the manuscript reports it. Its harness has been corrected — it previously used the all-to-all protocol the paper rejects — and its firing-rate penalty retuned, but it still needs a zero-recurrence control that separates from the connectomes before any number from it means anything. See `AUDIT.md` §D6.

Every number in the manuscript is emitted into `paper/generated_numbers.tex` by `generate_numbers.py`; none is transcribed by hand, and a missing result appears in the PDF as a visible `??` rather than as a stale value.

Tests

```
.venv/bin/pytest
```

The suite covers the census against brute-force enumeration, exactness of the incremental updates, degree and valence preservation in every null, convergence and randomisation of the annealed chain, the reservoir protocol controls, and the perturbation bounds.

Documents

File	What it is
info.md	**Everything about the project** — question, method, results, what is not cla
AUDIT.md	Every withdrawn claim and the evidence against it, including defects found in
pitch/EXECUTIVE_PITCH.md	Executive summary (v5 — replaces a withdrawn one)
paper/computational_coupling_v5.pdf	The manuscript
research/	Planning documents from the project's start, superseded; kept so the retraction
paper/superseded_v4/	The retracted v4 manuscript, figures and pitch dossier

A note on what is and is not in this repository. The retracted v4 manuscript, monograph, supplement and pitch dossier are kept as **LaTeX sources** under paper/superseded_v4/, so every claim `AUDIT.md` cites can be checked against what was actually written. Their compiled PDFs are not published here: their title pages carry the authors' contact details, and `build_docs.sh` already refuses to render fresh copies of withdrawn claims (see its header). The retraction notices in those directories are tracked and rendered.

A PDF of each Markdown document is built alongside it by `./src/analysis/build_docs.sh`, which first runs `check_doc_numbers.py` to verify that every figure quoted in `info.md` and the pitch still matches `paper/generated_numbers.tex`. Those two documents are written for people rather than compiled from macros, so they are the one place a number could go stale against a re-run; the check refuses to render a PDF from prose that disagrees with the results. The retracted v4 pitch and manuscript are deliberately **not** rendered — their retraction notices are.

Citing

Machine-readable metadata is in CITATION.cff; GitHub renders it as the **"Cite this repository"** button in the sidebar.

```bibtex
@misc{pasala2026coupling,
  title  = {Computational Coupling: Four Choices Decide Whether a Connectome
            Beats Its Rewiring},
  author = {Pasala, Ashok Raj and Gorai, Snigdha and P., Jeevana Jyothi},
  year   = {2026},
  note   = {School of Computer Science and Engineering, VIT-AP University,
            Amaravati 522237, Andhra Pradesh, India},
  howpublished = {\url{https://github.com/Ashok-clg/computational-coupling}}
}
```

If you cite the null-ensemble result, please say **which** ensemble you mean — and cite the code that generates it, not a description of it. That is the point of the paper and of this repository: the phrase "degree-preserving randomisation" does not determine one.

License

MIT — see LICENSE.